Unit: 6 Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ____________________
Date: ____________

Challenge (Habanero)

Q1. Simplify $\sqrt{16} + \sqrt{9}$
(A) 5
(B) 7
(C) 4
(D) 3
(E) 6

Q2. What is $\sqrt{4} \cdot \sqrt{9}$?
(A) 6
(B) 12
(C) 5
(D) 3
(E) 2

Q3. In a gaming tournament, the score is calculated by $\sqrt{points}$. If Player A has 16 points and Player B has 25 points, what is the difference in their scores?
(A) 1
(B) 3
(C) 2
(D) 4
(E) 5

Q4. Simplify $(\sqrt{8} + \sqrt{2}) - (\sqrt{4} - \sqrt{2})$
(A) 4
(B) 2
(C) 3
(D) 6
(E) 5

Q5. A coder uses $\sqrt{256}$ bytes to store data. How many bytes does she use?
(A) 16
(B) 8
(C) 12
(D) 10
(E) 14

Q6. Simplify $\frac{\sqrt{48}}{\sqrt{3}}$
(A) 4
(B) 2
(C) 6
(D) 8
(E) 12

Q7. In a game, $\sqrt{x}$ represents the number of lives. If $x = 64$, how many lives does the player have?
(A) 8
(B) 6
(C) 10
(D) 4
(E) 12

Q8. Simplify $(\sqrt{3} + \sqrt{2})^2$
(A) 7
(B) 5
(C) 9
(D) 3
(E) 11

Open Ended Questions (FRQ)

Q9. A data analyst has two sets of data, each represented by a radical expression. Set A is $\sqrt{50}$ and Set B is $\sqrt{18}$. Simplify the sum of the two sets and show your work.

Q10. A gamer's score is represented by the product of two radicals, $\sqrt{32} \cdot \sqrt{2}$. Simplify the expression to find the gamer's score and explain your reasoning.

Chef's Tips

- Emphasize the importance of simplifying radicals
- Encourage students to use real-world examples to illustrate their answers
- Remind students to show all work and calculations for free-response questions